Systems Analysis & Design

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Systems Analysis - Local business support recovery

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System Overview and Scope

System Description

The system analyzed is Titan Plaza shopping mall, located on Av. Boyacá with Calle 80, one of the most visited shopping malls in Bogotá. It was selected as a case study for the "Local Business Recovery Support System" project, which aims to develop an analytics platform to support small businesses' post-pandemic recovery through customer engagement and inventory optimization.
Although Titan Plaza houses large chains such as Falabella and Éxito, it also contains numerous small independent businesses that are the focus of this analysis. The shopping mall was chosen because it concentrates a wide variety of small businesses in one place, facilitating data collection and allowing observation of how these shops face challenges related to customer traffic, inventory turnover, and loyalty in a real world environment.
This analysis of Titan Plaza will provide real world insights into small businesses' challenges, serving as the foundation for designing the Local Business Recovery Support System in subsequent workshops.

Main Components and Boundaries

Titan Plaza has 4 floors dedicated to commerce, 250 stores, a food court, a cinema, department stores such as Falabella, and more than 2,000 parking spaces. The shopping mall is open daily from 8:00 a.m. to 10:00 p.m.
Within the scope of this analysis are the small independent businesses, their employees, customers, and common areas such as hallways, rest areas, and food courts.
Excluded from the scope are large chain stores, the mall's administrative offices, loading and unloading areas, external suppliers, and stores located in the surroundings of the shopping mall.

Stakeholders

- Customers: People who visit the mall to shop, eat, or spend time with family. They look for good prices, variety, and a pleasant experience.
- Employees: They assist customers and manage inventory daily. They know firsthand how work flows and what customers need.
- Store Owners: Entrepreneurs or managers responsible for sales, inventory, and profitability. They seek to adapt to new shopping habits.
- Mall Administrators: They manage common areas, security, and overall mall operations. Their goal is to keep the mall attractive and busy.
- Suppliers: They provide the products sold in stores. Their relationship with store owners is key to keeping inventory available.

Context

This analysis takes place in the post-pandemic context (2024-2026), where consumer behavior has significantly shifted. Studies show that customers now prioritize value for money, prefer open and ventilated spaces, combine online and physical shopping, and seek experiential retail over mere product acquisition (McKinsey, 2023).
Titan Plaza is located in northwestern Bogotá, attracting residents from middle-income neighborhoods (strata 3-5).
This analysis will identify real needs of small merchants regarding customer engagement and inventory optimization, serving as foundation for the proposed system.

Data Collection Plan and Execution

Methodology Design

- **Direct Observation:**

  Through direct observation, the flow of people entering and exiting the shopping center was analyzed, as well as the behavior of visitors within its different areas. During the observation, the areas with the highest concentration of people and the activities carried out by users in specific locations were recorded.

  From this observation, it was identified that some of the busiest areas are the cafes, especially Starbucks and Tostao, as well as the gym. This may be because many people use these spaces not only to purchase products, but also to work, study, or meet. Additionally, it was observed that Sundays tend to be days with a greater presence of families.

- **Surveys:**

  Information was collected from employees and customers who frequent the shopping center using a survey method. The questions focused on identifying the areas they visit most often within the mall and how they perceive the flow of people and revenue before and after the pandemic.

  Based on the responses, some patterns were identified regarding visitor preferences and their perception of the mall's operations. The results allowed for a comparison of how foot traffic has changed since the pandemic and which spaces or activities are currently most frequented by users.

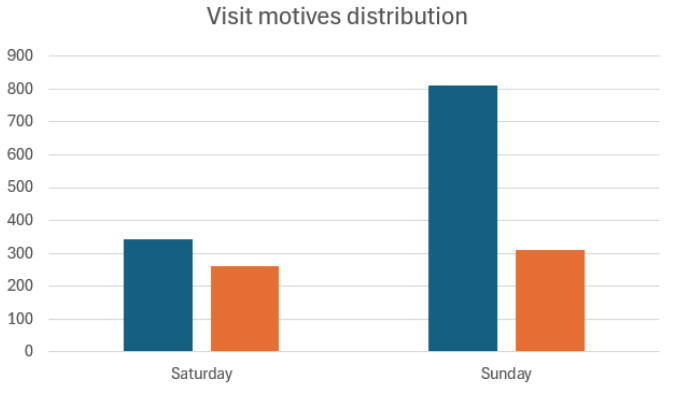
- **Environmental Correlation:**

  Using the environmental correlation method, different environmental factors that could influence the flow of people within the shopping center were analyzed. For this purpose, observation sessions were conducted on two different days (Saturday and Sunday), taking into account variables such as weather, time of the month, in-store promotions, and the type of visitors present.

Primary Data Presentation

- **Graphic 1:** Foot Traffic Comparison

ANSWER	ENTER	EXIT
Saturday (9am - 12pm)	343	262
Sunday (12pm - 3pm)	811	309
TOTAL	1154	571



- **Graphic 2:** Visit motives distribution

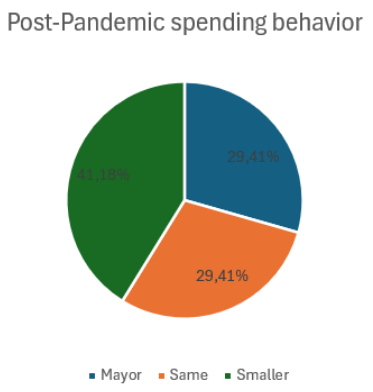
ANSWER	FREQUENCY	%
Shop in small / Local stores	2	11,76%
Food court / Restaurants	3	17,65%
Cinema / Entertainmen	3	17.65%
Large stores	2	11.76%
Just for a walk / Accompanying someone	2	11.76%
Other	5	29.41%
TOTAL	17	100%

JTION

rants

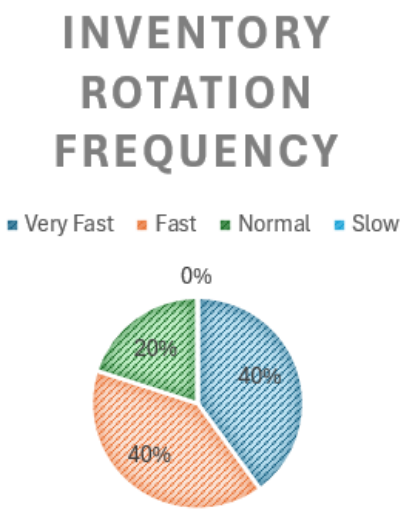
- **Graphic 3:** Post-Pandemic spending behavior

ANSWER	FREQUENCY	%
Mayor	5	29,41
Same	5	29,41
Smaller	7	41,18
TOTAL	17	100%



- **Graphic 4:** Inventory Rotation Frequency

ANSWER	FREQUENCY	%
Very Fast (less than a week)	2	40%
Fast (1 - 2 weeks)	2	40%
Normal (2 - 4 weeks)	1	20%
Slow (more than a month)	0	0%
TOTAL	5	100%



Data Quality Assessment

Although the data collected allowed us to obtain useful information about the behavior of visitors and the flow of people in the shopping center, it is important to consider some limitations of the collection process:

1. Firstly, the number of surveys carried out was relatively small, so the results represent only a sample of user behavior and do not necessarily reflect all visitors.

2. There may also be possible biases in the responses, since some people may have responded based on their personal perception or individual experiences, which may influence the objectivity of the results.

3. On the other hand, some external factors, such as the weather, promotions in stores or the specific day of the week, could influence the flow of people during the observation period.

4. Finally, some challenges arose during data collection, such as limited time to conduct observations and the availability of people willing to respond to the surveys.

Despite these limitations, the combination of several collection methods allowed obtaining an overview of the system behavior and improving the reliability of the analysis.

System Analysis

Element and Relationship Analysis

The local business recovery system is composed of several elements that interact with each other, such as local businesses, customers, business managers, online platforms, and local authorities.

Businesses offer products and services, while customers generate demand through consumption. Business managers implement strategies to adapt to market conditions, and online platforms facilitate the promotion and sale of products. Local authorities can influence the system through regulations or support measures.

The interaction between these elements creates a network of relationships in which changes in one component can affect the behavior of the system as a whole.

Sensitivity and Variability Study

The system is sensitive to changes in different factors, such as customer behavior, business strategies, economic conditions, weather, and the level of attendance at the shopping mall.

Temporal Variability (Weekday vs. Weekend): The system exhibits high variability based on the day of the week. Observing the data collected on Sundays, it shows a state shift where the customer flow changes from gym and co-working during weekdays to cinema and food Court. While foot traffic increases on weekends, the conversion rate for small businesses does not scale proportionally, as recreational users tend to prioritize entertainment over specialized local shopping. This behavior may be due to the fact that larger chain stores dominate this business area; meanwhile, small merchants are focused on the sale of products or services, rather than entertainment.

Sensitivity to Product Availability: A critical sensitivity point identified in the data is the lack of identification of customer needs to offer products accordingly. A large portion of surveyed customers reported being "Frequently" or "Sometimes" unable to find specific items in small stores. The system is highly sensitive to this gap; a single negative experience regarding availability triggers a loop where the customer permanently redirects their demand toward large anchor stores (Falabella/Éxito) or e-commerce platforms.

Influence Sensitivity: The decision-making process of the customer is highly sensitive to two main variables: Price and Recommendations. Survey results indicate that local businesses are vulnerable to price fluctuations, for example, the cost of leasing premises within the shopping center. Furthermore, a significant sensitivity to "Word-of-Mouth" was detected, meaning the system's stability depends heavily on informal information flows rather than structured marketing.

Complexity and Emergent Behavior

The small business ecosystem in Titan Plaza functions as an adaptive system where individual interactions between customers and local merchants give rise to patterns that cannot be explained by looking at the parts in isolation

Feedback Loops and System Complexity: Reinforcing Loop (Word-of-Mouth): A positive impact with a small business triggers a word-of-mouth effect. This activates a feedback system that directly increases the sales effectiveness of small merchants, as informal recommendations stabilize their position within the complex mall environment.

Balancing Loop: Large chain stores can diminish the potential customer base of small shops. If a small store fails to offer a unique value proposition or personalized attention, the system automatically redirects the customer flow toward the more "stable" large-scale nodes.

The complexity of the ecosystem is heightened by the lack of a unified data interface, where each small shop operates as an isolated agent with limited information regarding the overall customer flow patterns detected in our analysis. This lack of connectivity leads to suboptimal inventory decisions, a key emergent property of a fragmented system that becomes evident when compared to large retail chains; the latter, by possessing direct digital contact channels, allow customers to verify availability online and go directly to the physical store, eliminating the possibility of the user exploring or discovering small businesses during their visit. Consequently, it can be inferred that if local merchants had a similar direct contact medium, they could identify the most sought-after products and dynamically adapt their inventory to actual customer needs, mitigating the current informational disconnect within the system.

Findings and Pattern Recognition

From the data collected through direct observation, surveys and environmental assessment, several paterns in the behavior of visitors to the shopping center could be identified:

Direct observation allowed us to identify:

- Some areas have a greater concentration of people, especially cafes, restaurants and the gym
- These places are not only used to consume products, but also as places to work, study or meet.
- Generating a greater permanence of visitors in these areas

The surveys carried out with clients and employees allowed us to identify:

- Trends in the way people use the shopping center, many visitors indicated that they frequent the location primarily to eat visit shops or spend free time.
- Some employes noted that the flow of people has changed before and after the pandemic suggesting a variation in visiting and consumption patterns.

Environmental Correlation allowed us to identify:

- The flow of people varies depending on the day of the week and external conditions (such as weather and so on):
  For example, Sunday saw a considerably greater flow of visitors, mainly families coming to the shopping center for lunch or a walk.
- There are also some factors such as promotions in stores or weather conditions, such as rain, that can influence the increase in the number of visitors or decrease it.

Critical Assessment

System performance evaluation:

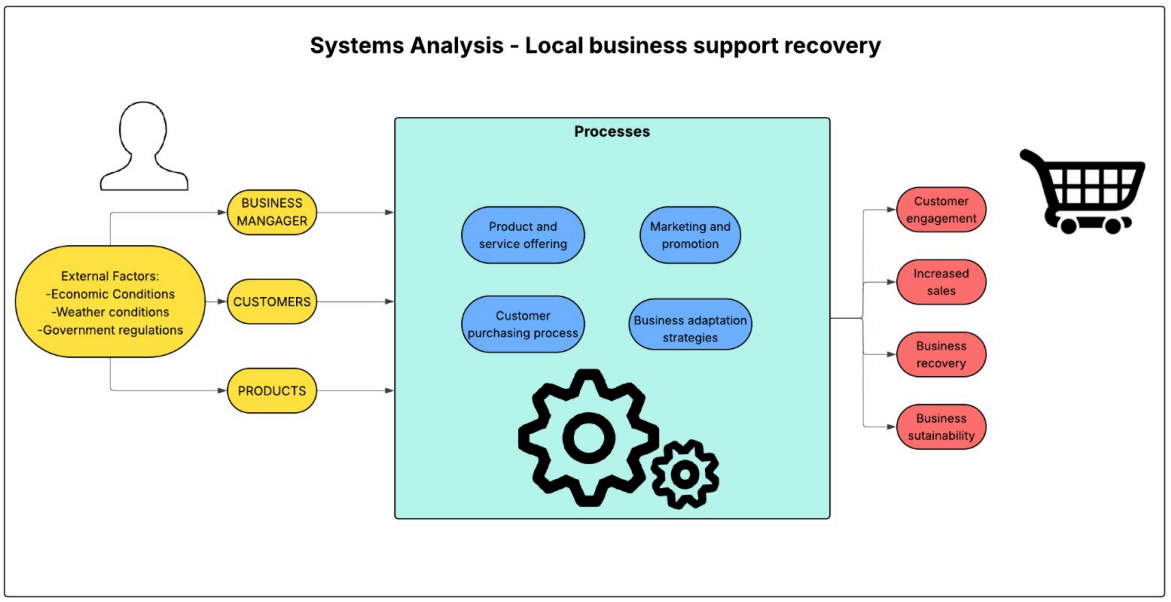
The system analysis of the small business ecosystem in Titan Plaza reveals a significant structural disconnect between potential demand and effective supply. Although the shopping center successfully attracts a massive flow of users through entertainment nodes such as gyms and cinemas, the system fails to transform this traffic into growth for small merchants due to a lack of informational integration.

Bottleneck identification:

A critical failure is the visibility gap; while large chains use digital interconnectedness to secure customer visits, small businesses operate as isolated agents with no real-time capacity to adapt their inventory to identified customer needs. This creates a fragile system where the small merchant is highly sensitive to external variables, such as leasing costs and price fluctuations, without possessing the digital tools to mitigate these risks.

Optimization opportunities and recommendations:
From a systems engineering perspective, the ecosystem requires a transition from a fragmented model to an integrated information network that allows local businesses to compete through data-driven adaptation, rather than relying solely on accidental pedestrian flow.

Visual System Representation



Requirements Analysis

Based on the findings from Workshop #1, several key requirements were identified to address the challenges faced by small businesses in Titan Plaza. The system must bridge the gap between customer demand and business supply, improve their visibility, and enable more informed decision-making through the use of data.

Functional Requirements

- Customer Traffic Information
  The system must inform users through surveys to other users about customer traffic, considering factors such as weather, day of the week, and time. A significant increase in traffic is observed depending on the day of the week and time.
- Inventory Alerts
  The system must notify owners when a product has low turnover or is running out. Although no store reported low inventory frequency, it is important that no store ever runs out of stock of any product.
- Automatic Promotions
  The system must send promotions. 41.18% of customers spend less now than before the pandemic. Promotions encourage them to buy and increase the visibility of many businesses that are not so visible in the shopping mall.
- Dashboard for Owners
  The system must display key metrics: sales, traffic, inventory turnover in real time. Owners need to make informed decisions.
- Product Recommendations
  The system must suggest products to customers based on previous purchases and trends. Improve conversion rate.

Non-Functional Requirements

- Security
  Customer and sales data must be encrypted and protected.
- Usability
  The interface must be intuitive for store owners; not everyone is a technology expert.
- Availability
  The system must have 99.5% uptime during business hours (8:00 a.m. to 10:00 p.m.).
- Scalability
  It must support growth to additional shopping malls without requiring a major redesign.
- Portability
  The system must work on mobile devices (Android/iOS) and web.

Constraints

- Data protection

  Must comply with Colombian personal data protection laws (Law 1581 of 2012).

- Owner distrust

  Not all store owners are willing to start using a system and share private information about their businesses.

- Connectivity

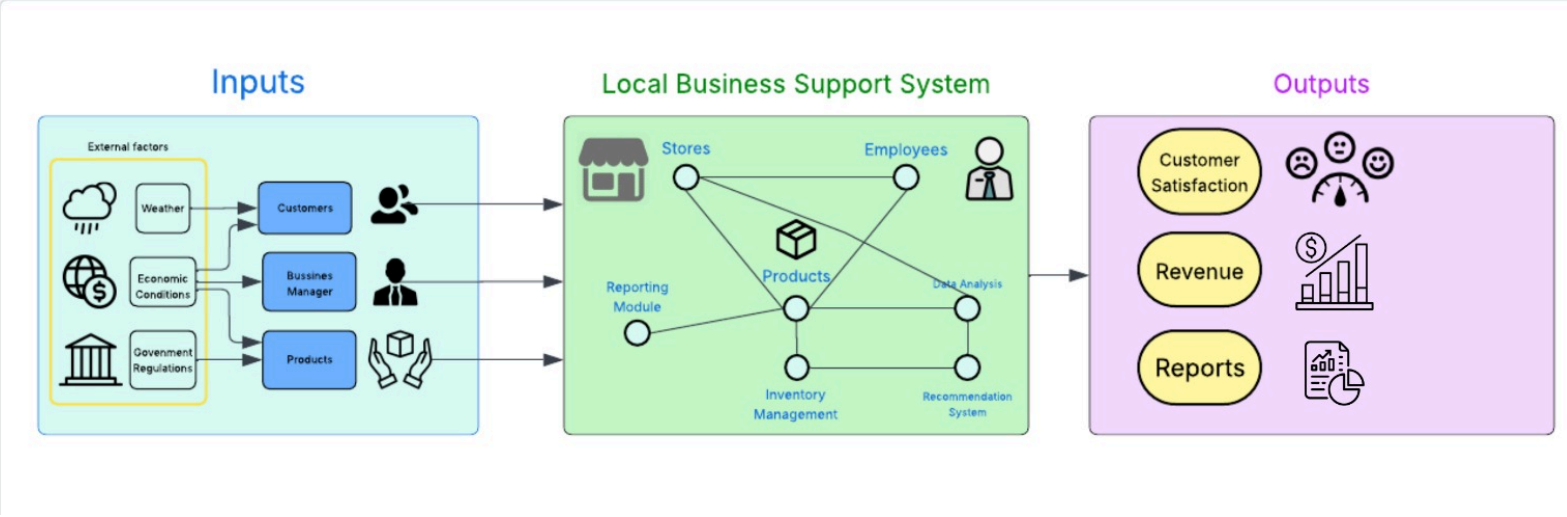
  Some areas of the mall have variable internet signal; must work partially offline.

- Existing systems

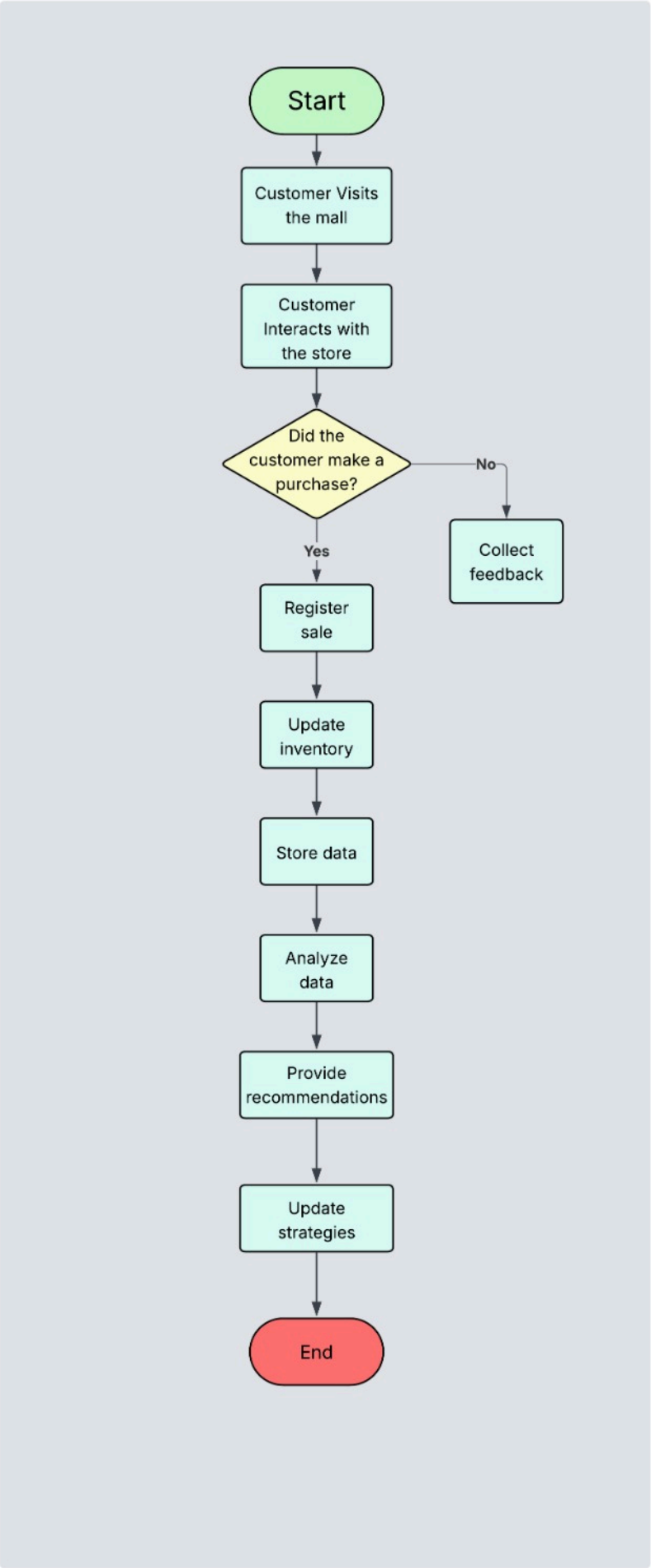
  Many owners already use dedicated inventory rotation systems; the solution must be able to integrate with them or coexist without causing conflicts.

System Architecture

System Architecture Diagram



PROCESS FLOW DIAGRAM



Architectural Overview

The system is based on a modular architecture centered on the **Local Business Support System**, which integrates data collection, processing, and visualization.

It consists of three main layers:

- **Data Input:** customer, business manager and products
- **Processing:** Local business support system
- **Output:** Revenues and reports for decision-making

This design is simple, scalable, and suitable for small business environments.

System Components

The main components of the system are:

- **Customers:** generate data through interactions and purchases
- **Business Owner / Manager:** makes decisions based on system outputs
- **Store Operations:** core business activities
- **Employees:** handle customer interactions and sales
- **Inventory Management:** controls stock levels
- **Sales Module:** records transactions
- **Data Analysis Module:** processes and analyzes data
- **Recommendation System:** suggests actions
- **Reporting Module:** presents information

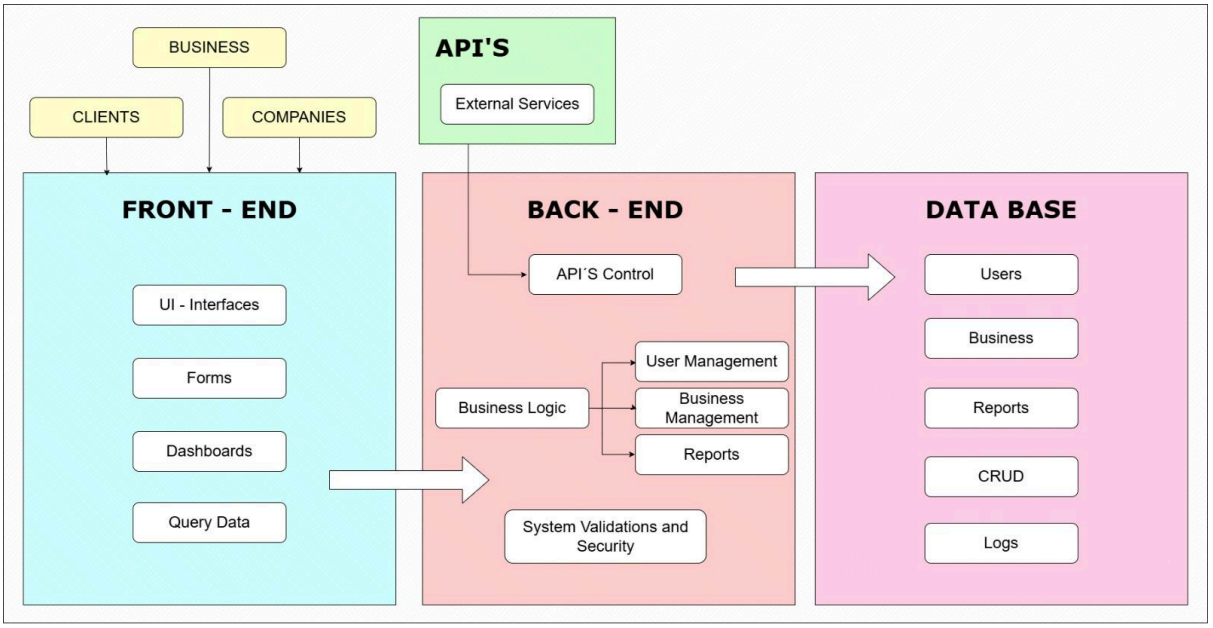
Data Flow Design

The process starts when customers interact with the store. If a purchase is made, the sale is recorded and inventory is updated. If not, feedback can be collected.

All data is stored and then analyzed to identify patterns such as demand trends and customer preferences. The system generates insights and recommendations, which are used by the business owner to adjust strategies.

This creates a continuous improvement cycle based on data.

Interfaces and Integration



Complexity and Risk Management

Identified Risks

About 41% of customers are spending less than before the pandemic, which makes it harder to keep them engaged with local businesses. Foot traffic varies a lot: Sundays bring in more than double the visitors compared to Saturdays, and weather also plays a big role, making it tough for store owners to manage inventory properly. Nearly 30% of visitors come to the mall for reasons other than shopping, so reaching them with promotions isn't easy. There's also the risk of running out of popular products if inventory alerts don't work well. On top of that, many store owners are reluctant to share their business data, and some areas of the mall have poor internet connection, which could affect how often people use the platform.

Mitigation Strategies

We'll send more personalized promotions, especially during peak traffic days. The dashboard will show traffic predictions so owners can prepare ahead and stock up on what sells most. Inventory alerts will notify them before products run out. We'll also provide hands-on training to help owners feel more comfortable with the system, and the interface will be kept simple. For areas with spotty internet, the platform will work offline and sync later. And for those already using other systems, we'll make sure everything can integrate smoothly.

Monitoring and Control Mechanisms

We'll track weekly how many owners are using the dashboard and whether they're satisfied with it. Promotion effectiveness will be measured by looking at sales increases and customer

response. Predictions will be compared to actual foot traffic to fine-tune the model. We'll keep an eye on inventory alerts to see if they actually prevent stockouts. Connection errors will be logged to improve offline performance, and regular audits will ensure customer and business data stays protected.

Technical Implementation Strategy

Technologies and Tools

The platform is structured across three main layers, as illustrated in the System Architecture Diagram. The front-end layer handles all user-facing components, including the business dashboard, data query interfaces, and reporting forms that store owners interact with directly. The back-end layer manages the core business logic, covering user management, business management, report generation, system validations, and security controls, as well as the integration with external services through API connections. Finally, the database layer stores all persistent information including user records, business data, reports, operation logs, and CRUD interactions. This three-layer separation ensures modularity and maintainability, allowing each component to be updated or scaled independently as the platform grows beyond Titan Plaza.

Development Methodology

The system will be developed following an iterative and incremental approach, where each cycle delivers a functional subset of the platform that can be tested and validated by store owners. This is particularly important given that a significant portion of business operators have limited familiarity with technology, meaning the interface and features must be progressively refined based on real user feedback rather than delivered all at once. Each iteration will prioritize the most critical functionalities first — such as the dashboard and inventory alerts — before advancing toward more complex features like demand forecasting and recommendation systems. This approach also reduces the risk of building components that do not align with actual operational needs, ensuring that the final solution remains practical, usable, and directly responsive to the challenges identified during the systems analysis of Titan Plaza.

Deployment Strategy

The system will be deployed in a controlled and gradual manner, beginning with a pilot phase involving a small group of volunteer stores within Titan Plaza. This initial rollout allows the team to validate the platform under real operational conditions, identify integration issues with existing business workflows, and build trust among store owners before broader adoption. During this phase, participating businesses will receive direct support to ensure a

smooth onboarding experience. Once the pilot results are validated and necessary adjustments are incorporated, the platform will be made available to all small businesses within the mall. Given that some areas of Titan Plaza present variable internet connectivity, the deployment process will include offline synchronization mechanisms that allow the system to continue operating under limited network conditions and sync data once connection is restored. As the platform matures and demonstrates measurable value, the architecture is designed to support expansion to additional shopping malls with minimal reconfiguration, treating each new location as an independent operational unit within the same infrastructure.

Implementation Phases

The development and deployment of the platform will be structured across four sequential phases. The first phase focuses on the foundational setup of the system, including the definition of the database schema, the core business logic, and the basic structure of the owner dashboard with read-only metrics. The second phase introduces the active modules of the platform, particularly the sales recording system, the inventory management module, and the automated alert mechanism that notifies owners when stock levels are critical. The third phase incorporates the analytical layer, integrating the demand forecasting model and the product recommendation engine built upon the traffic and behavioral patterns collected during Workshop #1. The fourth and final phase covers the pilot deployment with selected stores, the collection of user feedback, performance tuning, and the full rollout to all small businesses within Titan Plaza, accompanied by training sessions to ensure proper adoption across the different levels of technological familiarity among store owners.

Performance and Optimization

Performance Metrics

The system's performance will be evaluated through a set of measurable indicators directly tied to the challenges identified in Workshop #1. Sales conversion rate will be tracked to determine whether the platform effectively translates visitor traffic into purchases for small businesses. Inventory stockout frequency will be monitored to assess whether the alert system successfully prevents product unavailability. Dashboard response time will be measured to ensure that store owners receive real-time data without significant delays. Additionally, promotion engagement rate will be tracked to evaluate whether the automated promotional features generate a measurable increase in customer interaction and spending, particularly relevant given that 41.18% of surveyed customers reported spending less than before the pandemic.

Optimization Strategies

The system will implement caching mechanisms for frequently accessed data such as current traffic estimates and top-selling products, reducing server load and improving dashboard responsiveness. The demand forecasting model will be continuously retrained as new sales and traffic data is collected, progressively improving the accuracy of its predictions over time. Promotions will be dynamically adjusted based on observed customer behavior patterns, ensuring that incentives are relevant and timely rather than static. For areas of the mall with poor connectivity, data synchronization will be optimized to prioritize critical updates — such as inventory changes and sales records — over less time-sensitive information.

Scalability Considerations

The platform is designed to scale horizontally as the number of participating stores grows, both within Titan Plaza and across future mall locations. Each shopping mall will operate as an independent unit within the shared infrastructure, ensuring that the growth of one location does not affect the performance of others. The analytical component will be deployed as a separate service, allowing it to be scaled independently from the rest of the system during periods of high data processing demand, such as weekends when foot traffic at Titan Plaza reaches its peak.

Conclusion and Future Work

The design presented in this document represents a direct response to the structural challenges identified during the systems analysis of Titan Plaza in Workshop #1. The core finding of that analysis — that small local businesses suffer from an informational disconnect that prevents them from competing effectively with large retail chains — has been addressed through the design of an analytical platform that centralizes data, automates critical alerts, and enables data-driven decision-making for store owners who previously relied solely on intuition and informal information flows.

The proposed architecture balances technical robustness with practical feasibility, taking into account the real constraints of the environment: variable internet connectivity, limited technological familiarity among store owners, existing inventory systems that must be respected, and the legal obligations imposed by Colombian personal data protection regulations. Rather than proposing an overly complex solution, the design prioritizes usability and gradual adoption as the primary drivers of long-term success.

Looking ahead, several opportunities exist for the continued evolution of the platform. As more sales and traffic data is accumulated over time, the accuracy of the demand

forecasting and recommendation models will improve significantly, generating increasingly precise insights for store owners. Future development cycles could also explore the integration of loyalty programs and direct customer-facing features, transforming the platform from a purely internal business tool into a broader ecosystem that connects customers and merchants within the mall. Finally, the multi-mall expansion model opens the possibility of generating cross-location benchmarks, allowing store owners to compare their performance against similar businesses in other shopping centers and identify further optimization opportunities.

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